

Definition Generation and Gloss-Grounded Word Sense Disambiguation for Sanskrit: A Reproducible Evaluation on the Dictionary Side of the DCS–CDSL Crosswalk

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Abstract

Large language models are increasingly evaluated on lexicographic tasks—generating dictionary-style definitions and selecting word senses—but Sanskrit, despite its exceptionally rich lexicographic tradition, has been largely absent from this line of work. The one published Sanskrit WSD-from-definitions system operates on the corpus side of the Digital Corpus of Sanskrit (DCS). We present the first evaluation of this task class on the *dictionary* side: definition generation and gloss-grounded sense selection scored against Monier-Williams and Heritage (Huet) glosses, connected to corpus attestations through the DCS–CDSL headword crosswalk. All data are local, committed and license-checked. Our definition-generation track evaluates five system arms on a frozen, stratified 500-headword sample with deterministic metrics and a blinded LLM judge under explicit gates; a cross-lingual re-scoring against Huet’s French glosses shows the system ranking to be reference-invariant. Our WSD track contributes a fully deterministic gold—DCS word-sense annotations resolved onto MW sense divisions—and shows that a gloss-grounded LLM arm (54.3%) falls far short of a corpus-frequency baseline (84.3%) on held-out polysemous items, a negative result that locates the difficulty of Sanskrit WSD in sense granularity and usage priors rather than in gloss access. Every number in this paper reproduces from committed artifacts with one command per track.

[Author-voice pass: Abstract drafted by the experiment lane; the thesis sentence and the framing of the negative result are the author’s to own. Gate: MG abstract approval (GTD ISCLS-9, due 10-10-2026).]

1 Introduction

Computational lexicography for Sanskrit sits on two large, independently curated resources: the dictionary tradition—above all Monier-Williams (Monier-Williams, 1899) and Huet’s Sanskrit Heritage Dictionary (Huet, 1994 2026)—and the Digital Corpus of Sanskrit (DCS) (Hellwig, 2010 2026), a 5.7-million-token morphologically annotated corpus whose recent releases carry human word-sense annotations over 531,747 tokens. A growing “definition modeling” line of work evaluates language models on generating and using dictionary definitions (Periti et al., 2025; Liu et al., 2026), and Hellwig et al. (Hellwig et al., 2026) have shown that lexicographic definitions can drive word sense disambiguation for Sanskrit at scale.

That system, however, lives entirely on the corpus side of the DCS–CDSL headword crosswalk. The dictionary side—where the glosses are the object of evaluation rather than a training signal—remains unevaluated: no definition-generation or gloss-grounded WSD evaluation has been run over the CDSL dictionaries, and none over the crosswalk that links them to corpus evidence. This paper closes that gap with two tracks over one frozen headword universe.

Track 1: definition generation. Five system arms—a derangement floor, three API base-lines of two model families, and a strong-model reference—generate English glosses for a frozen, frequency- and polysemy-stratified sample of 500 Monier-Williams headwords, scored with chrF, token-overlap F1 and a blinded LLM judge under explicit gates. A second reference re-scores the same arms against Huet’s French glosses on the 333-headword $MW \cap \text{Heritage}$ subset.

Track 2: gloss-grounded WSD with a deterministic gold. For polysemous lemmas of the same universe we build a sense-selection task whose gold requires no manual annotation: a DCS sentence, the lemma’s numbered MW sense glosses, and the correct sense recovered by joining DCS’s own word-sense annotations to MW sense divisions through the crosswalk. We score a gloss-grounded LLM arm against the corpus-frequency (MFS) baseline that the same data support, with paired significance.

Our contributions are: (i) the first definition-generation evaluation on Sanskrit lexicographic data, with its cross-lingual reference-invariance check; (ii) the first gold-scored gloss-grounded WSD evaluation on the dictionary side of the crosswalk, built without any new annotation; (iii) an honest negative result—prompting with glosses and context is not competitive with corpus priors on this gold—and the error analysis that localises it; and (iv) a fully reproducible evaluation stack (frozen samples, committed artifacts, one-command re-scoring and validation).

2 Related Work

Definition modeling. Periti et al. (Periti et al., 2025) survey definition generation for word meaning modeling across monolingual, multilingual and cross-lingual settings; LM-Lexicon (Liu et al., 2026) improves definition modeling by harmonising semantic experts. Neither line touches Sanskrit. Our Track 1 imports their evaluation class (reference-based surface metrics plus a judged adequacy score) into Sanskrit lexicography.

Sanskrit WSD. Hellwig et al. (Hellwig et al., 2026) train a supervised bi-encoder on MW lexicographic definitions and evaluate on DCS-anchored gold, establishing that definitions are a usable sense signal for Sanskrit. Patel and Kulkarni (Patel and Kulkarni, 2024) align senses *across* Sanskrit lexica—the operation our gold construction depends on, here used deterministically via the DCS–CDSL crosswalk rather than as a learned model. Our Track 2 differs from (Hellwig et al., 2026) in three deliberate ways: it is zero-shot prompting rather than a fine-tuned system, it scores on the dictionary side of the crosswalk, and it reports against a corpus-frequency baseline with paired significance, accepting in advance that a negative outcome is a publishable finding.

LLM judges. We use an LLM judge only under the standard gates—floor separation and correlation with a deterministic metric—motivated by large-scale evidence that LLM judgments do not automatically match human ones (Bavaresco et al., 2025). A human-scored subsample is scheduled before camera-ready and flagged in Limitations.

3 Data

All datasets are pre-existing, committed assets consumed, not rebuilt.

A3: MW glosses. 187,506 SLP1-keyed English glosses extracted from Monier- Williams (Monier-Williams, 1899) (public domain, 1899). Glosses carry markup- stripping artifacts; all arms share them equally.

D20: Heritage glosses. 24,549 entries from Huet’s Sanskrit Heritage Dictionary (Huet, 1994 2026), keyed by MW headword with entry anchors; used as the independent second reference.

C15: the crosswalk and the sense map. The DCS↔CDSL headword crosswalk (98,606 rows) links corpus lemmas to dictionary entries and defines the frozen universe’s membership. A companion committed map resolves DCS word-sense synset ids onto MW sense divisions for the WSD gold; its exact/overlap rows cover 32,369 (synset, lemma) pairs.

DCS 2026. The Digital Corpus of Sanskrit (Hellwig, 2010 2026): 5.69M tokens, 754,726 sentences, with per-token word-sense annotations (`m_wordsem`) over 531,747 tokens, and a lemma inventory with IAST lemmas and grammar tags.

E26: corpus frequency. Lemma frequency sidecar (83,277 DCS lemmas) used for the frozen sample’s frequency strata.

Licenses: MW gloss text is public domain; DCS text and annotations are CC BY with attribution to Oliver Hellwig; the frozen samples and gold tables are released under the kosha public-tier license with those attributions. No external data were added.

4 Evaluation Design

4.1 Frozen headword universe

The two tracks share one frozen 500-headword sample (seed 730, deterministic builder, SHA-256 digests of all inputs committed): the universe is the 11,316 SLP1 keys present in $A3 \cap E26 \cap C15$, stratified 3×3 by corpus frequency rank (high/mid/low) and MW top-level sense count (mono / 2–4 / ≥ 5), 56 cells targeted, final $n = 500$ (one cell exhausts at 52; every skip is logged). A candidate requires at least three usable DCS attestation sentences (4–40 tokens, ≤ 2 per work). The freeze is the committed sample plus attestation files; scoring never re-derives the join.

4.2 Track 1: definition generation

Each arm produces an English gloss for a headword given its grammar tag and up to five DCS attestation sentences (context arms), or the headword and tag alone (the no-context arm):

- **A0 (floor):** a seeded derangement of the gold glosses—metric floor and judge-discrimination control.
- **A1 / A2: deepseek-chat** (DeepSeek-AI, 2025) with and without attestations (temperature 0). The A1–A2 delta isolates what corpus context adds over parametric memory.
- **A3: deepseek-reasoner** with attestations.
- **F1 (reference):** a stronger non-DeepSeek frontier model, same input contract, generated in-session under a gold-free inputs projection; treated as a reference point, not a reproducible baseline arm.

Scoring combines deterministic metrics (corpus and sentence BLEU and chrF via sacrebleu (Post, 2018); bag-of-tokens F1) with a blinded LLM judge (**deepseek-chat**, adequacy 0–5 vs. gold, blind to arm) admitted only under gates: the judge must separate the A0 floor and correlate positively (Spearman ρ) with chrF within system arms. Because MW 1899 is certainly in every candidate model’s pretraining data, the track measures *reproduction and grounding*, not generation from corpus evidence alone; the A1–A2 contrast and this framing carry the caveat explicitly.

The second-reference study re-scores all five arms on the 333-headword $MW \cap \text{Heritage}$ subset against Huet’s French glosses with a French-prompted blinded judge, asking whether the arm ranking survives a change of reference dictionary and language.

4.3 Track 2: gloss-grounded WSD with a deterministic gold

Item and gold. An item is one DCS sentence $\times$ one lemma of the frozen universe, held out by a pinned deterministic fold: $\text{crc32}(\text{sent_id}) \bmod 5 = 0$ ($\approx 20\%$). The gold sense is obtained with no manual step: the sentence’s `m_wordsem` synset id and the lemma, joined through the committed WordSem→MW sense map, yield an MW sense `lemma#ord`. The candidate inventory is the lemma’s attested MW senses from the corpus-derived sense-frequency table. Items enter the pool only if the lemma has 2–15 attested senses and the gold lies inside the inventory; monosemous items are excluded by design. The pool funnel over the frozen universe: 531,747 tagged tokens $\rightarrow$ 26,570 gold-mapped $\rightarrow$ 5,115 held-out items $\rightarrow$ 4,100 eligible polysemous $\rightarrow$ 300 sampled (seed 5877; 82 lemmas; inventory size k : min 2, median 3, max 12). In only 33.7% of items is the gold sense also the inventory’s rank-1 sense, so the sample is not trivially MFS.

Arms. **LLM: deepseek-chat** (temperature 0) receives the headword with grammar, the IAST context sentence, and the numbered sense glosses, and returns a JSON sense index. **MFS:** the training-fold majority sense per lemma (the corpus-frequency prior the gold annotations themselves support). **Floor:** uniform $1/k$. All arms are scored on identical items; the LLM and MFS comparison is paired and evaluated with the exact McNemar test.

Fold integrity. An earlier iteration of the gold spine keyed its train/test split on Python’s per-process-randomised `hash()`; its published split is stable in expectation but not reproducible byte-for-byte. This track pins the split (`crc32`) and re-derives its baseline under the pinned fold; we report the discrepancy openly rather than silently redefining prior artifacts (Section 5.3).

5 Results

5.1 Track 1: definition generation

All 500 items were generated and judged in every arm with zero empty candidates.

Arm	BLEU	chrF	token-F1	judge 0-5	$\rho(\text{judge}, \text{chrF})$
A0 floor	3.13	12.02	0.097	0.19	0.117
A1 chat + ctx	3.05	19.57	0.297	4.32	0.391
A2 chat no-ctx	3.43	17.30	0.294	4.08	0.466
A3 reasoner + ctx	0.50	13.70	0.256	4.21	0.423
F1 frontier + ctx	5.14	24.35	0.340	4.60	0.415

Table 1: Definition generation, 500-headword frozen sample. Judge gates pass in every arm: the floor is cleanly separated and judge–chrF correlation is positive.

Three findings. **(1) BLEU is uninformative here:** the register-matched floor matches the systems (3.13 vs 3.05), while chrF and token-F1 separate them cleanly. **(2) Context adds little over memory:** A1–A2 = +2.3 chrF, +0.003 F1, +0.24 judge—attestations sharpen sense selection, not content, consistent with MW being in pretraining data. **(3) Metric choice changes the ranking:** the terse reasoner is last on chrF (recall against 41.5-word golds) yet mid-pack under the blinded judge. The frontier reference F1 leads every metric simultaneously (Table 1), with judge–chrF agreement inside the healthy band—the surface/judge ranking concord the DeepSeek arms lack.

The polysemy penalty replicates in every frequency band (polysemy ≥ 5 is the worst cell everywhere), and the frequency gradient is inverted—low-frequency cells score *higher*—because high-frequency MW entries carry long, structurally complex gold glosses: a stratification effect, not model skill at rare words.

Second reference (Heritage, French). On the 333-headword $\text{MW} \cap \text{Heritage}$ subset the arm ranking is **reference-invariant:** $\text{F1} > \text{A1} > \text{A3} > \text{A2} > \text{A0}$ under both the English and the French-prompted judge. The MW-familiarity premium—the judge’s excess kindness to glosses close to MW’s own wording—is real but small (+0.13 to +0.25 on 0–5; bootstrap CIs exclude zero for every system arm; largest for the memorisation arm A2, smallest for F1). Cross-lingual surface metrics are near-degenerate; the judged score is the measurement there.

5.2 Track 2: gloss-grounded WSD

Arm	Accuracy	n
LLM (chat, gloss-grounded)	0.543	300
MFS (corpus prior)	0.843	300
Uniform floor	0.319	300

Table 2: Gloss-grounded WSD on held-out WordSem gold (300 polysemous items, pinned fold). Paired LLM vs MFS: 10 vs 100 discordant pairs, exact McNemar $p < 0.0001$.

The result is a clean negative (Table 2): the gloss-grounded LLM arm scores 54.3% against the MFS baseline’s 84.3% on identical items—30 points below, and only 22 points above the uniform floor. Zero API failures; all 300 picks parsed.

The gap is concentrated, not uniform. By inventory size: at $k = 2$ the LLM reaches 0.700 (MFS 0.940); at $k \geq 6$ it collapses to 0.183 (MFS 0.676). MFS degrades gracefully with k ; prompted gloss-matching does not. Three error classes dominate manual inspection of wrong items: *fine-granularity near-misses* inside one gloss family (*pitr*:- “father and mother (du.)” vs “the Fathers, ancestors”); *literal-vs-metaphorical* splits (*phala*:- “fruit” vs “fruit = retribution”); and *cross-POS homonymy* (*mad*:- pronoun base vs “to gladden”).

5.3 Fold integrity

The pinned-fold MFS baseline (0.843, polysemous-only, this sample) sits where the earlier unpinned pipeline’s published number lands (0.840 over all mapped test tokens, monosemous included): the randomised split moved the estimate by less than a point but made byte-level reproduction impossible. We therefore pin the fold, re-derive the baseline, and flag the earlier artifacts as computed-under-one-process-seed rather than rewriting them.

6 Discussion

[Author-voice pass: Interpretation slot — the author’s reading of the negative result goes here; the lane offers the material reading only. The gold’s corpus grounding means MFS exploits precisely the usage prior that the annotation encodes, while a single-sentence gloss match cannot; the supervised-encoder path of Hellwig et al. remains the strong way to consume definitions for WSD, and a frequency-aware or hybrid prompt is the obvious next ablation. Whether the paper leads with the resource (reference-invariant eval + deterministic gold) or with the negative methodological finding is an author’s decision.]

The two tracks read together as a statement about where the difficulty of Sanskrit computational lexicography lives. Definition content is largely recoverable from parametric memory (Track 1, $A1 \approx A2$), reference dictionary choice barely moves a system ranking, and the real discriminative work happens at sense granularity—exactly where Track 2 shows prompted models failing against corpus priors. For builders of Sanskrit dictionary services the practical implication is that sense-level services should lean on corpus-frequency priors first and treat gloss matching as a refinement signal, not the reverse.

7 Limitations

Judge human subsample. The blinded judge passes its gates and agrees with surface metrics on the frontier arm, but no human-scored subsample exists yet; a human-scored subset is scheduled before camera-ready and no judge-only claim stands alone until then. **Contamination.** MW 1899 is in pretraining data for every arm; Track 1 claims are framed as reproduction+grounding throughout. **Single scoring sample.** Track 2 reports one 300-item sample of a 4,100-item eligible pool; all artifacts to scale it are committed, and the pinned fold makes any rescaling reproducible. **Sandhi.** Context sentences are given in sandhied IAST as DCS stores them; unsandhied variants exist in the dump and their effect is unmeasured here. **Coverage.** The gold map covers exact/overlap crosswalk pairs; unresolved pairs are excluded from scoring, and the funnel counts above make the exclusion visible. **F1 arm.** The frontier reference arm is in-session and not temperature-pinned; it is labelled a reference, not a baseline.

GenAI acknowledgment

Per the ISCLS-9 CFP generative-AI policy we disclose: AI agents performed the experiments, implemented the evaluation harness, computed all reported numbers, prepared tables and formatting, and drafted structural prose. Substantive wording—the thesis, the interpretation of results, and the conclusions—passes through a human author-voice revision before submission, and the submitting author takes full responsibility for the published text. All numbers derive from committed, one-command reproducible artifacts within the authors’ own data estate.

Reproducibility

Every number in this paper regenerates from committed artifacts. Track 1: sample builder, five generation arms, scorer and judge scripts with the frozen sample and SHA-256 digests of all inputs. Track 2: `build` (deterministic item sample, seed 5877), `run-llm`, `score`, `validate`—the last recomputes every committed score and prints PASS only on an exact match. Data statements and licenses ship beside each dataset.

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